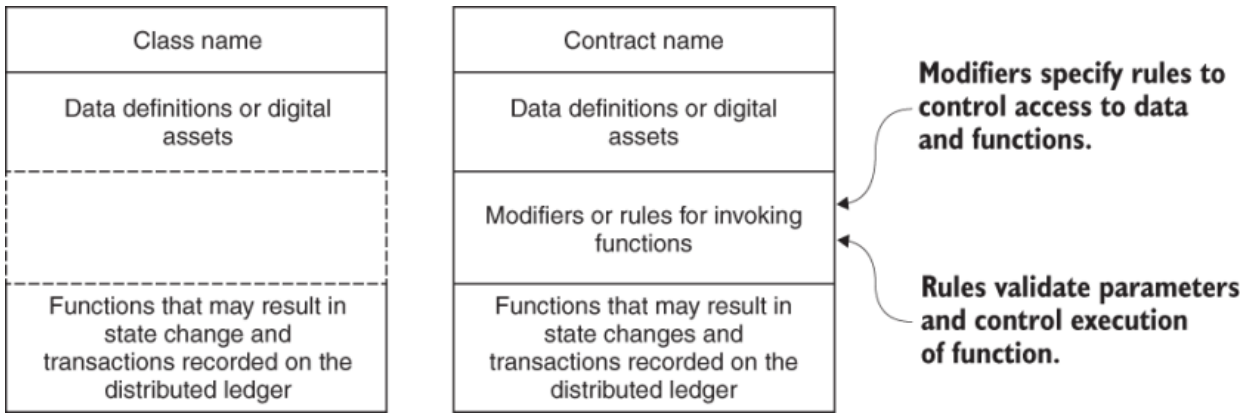


Figure 2.4 Class diagram versus contract diagram templates

The smart contract diagram on the right side of figure 2.4 has one additional component: the rules for accessing the functions and data. This component distinguishes a smart contract diagram from a traditional class diagram. Design principle 4 deals with contract diagrams.

> **DESIGN PRINCIPLE 4** Define a contract diagram that specifies the name, data assets, functions, and rules for execution of functions and access to the data.

In the simple use case of the counter, no conditions or rules are used. That's fine; we don't need any rules, because this problem is a simple one used to illustrate the design process. The contract diagram for the counter is shown in figure 2.5. You'll use the camel-case convention for the identifiers of the various components (data variables and functions) of the contract. The diagram shows the name of the contract as Counter, the only data element as an integer called `value`, and the functions from the use case diagram in figure 2.3, which is replicated on the left side of figure 2.5. Besides the three functions--`initialize()`, `increment()`, and